

ProofSettle

A settlement rail that pays for off-chain AI compute only when **two independent proofs, of two different kinds, agree inside a single transaction.**

Live on Creditcoin CC3 testnet

Real Attestcoin proofs

Real TEE, Google Confidential Space, AMD SEV

Verifiable in a browser at proofsettle.pages.dev



Amara prices loans for people with no credit file

She runs risk at a thirty-person lender in Lagos making working capital loans to small traders. **Two thirds of her applicants have no credit history at all.**

The only way to price them is to run a model over what they do have: mobile money records, supplier invoices, till takings. She has no GPUs and no ML team, so she would have to buy that inference from someone else.

She cannot. It is her borrowers' data, she is regulated on it, and a provider's promise not to keep a copy is only a promise. And if she bought it anyway, she would be paying up front for a result she has no way to check.

So she does not. The loans do not get made.

What she needs

To name the exact code she is willing to trust, at the moment she pays, and have that enforced by something other than a supplier's word.

The other side of the rail

The provider selling her that compute has the mirror problem: today they cannot get paid without asking a stranger to trust them first.

Throughout: an **enclave** is a sealed compartment on a server that can prove which code is running inside it, and which nobody, including the machine's owner, can see into.

THE PROBLEM

Two facts have to be true, and they live on different systems

Did the payment really happen?

It happened on another chain. The contract that would release the work cannot see it, so somebody has to be trusted to say so.

Did the right code really run?

It ran off chain, in someone else's data centre. The buyer gets an answer and a bill, and no way to tell which model produced either.

Solve one and you still have the other. **Most designs solve one**, then close the gap with a trusted coordinator, which is the thing everyone was trying to remove.

Both usual fixes replace a cryptographic question with a reputational one: trust the provider, or trust an oracle operator who watched them.

HOW IT WORKS

Compose two proofs, settle on the pair

In plain terms: **the money is held by a contract, not by either party**, and it only moves when two separate pieces of mathematics agree that the buyer paid and that the agreed code is what produced the answer.

ETHEREUM SEPOLIA

Amara pays, and names the enclave build she will accept

The requirement travels inside the payment event. The applicant's record rides with it, sealed to the enclave's key.

OFF CHAIN

A TEE opens the record, runs the named model, signs the verdict

Google Confidential Space, AMD SEV. Neither key leaves the enclave. The answer goes back sealed to Amara alone.

ATTESTCOIN ORACLE

Proves the Sepolia payment to Creditcoin

Merkle inclusion plus continuity. No oracle operator.

CREDITCOIN

Both proofs verified in one transaction, then money moves

Accepted pays the provider, rejected refunds, partial splits.

The proven foreign event does not just carry a fact. **It carries the buyer's policy**, and the contract on Creditcoin enforces that policy against a second, unrelated cryptographic system.

Neither half settles alone.

Most integrations ask the oracle for a fact

- | | | |
|---|---|--------------------|
| 1 | Call the prover, display a proof. The oracle is a widget. | SHALLOW |
| 2 | Burn on Sepolia, prove, mint on Creditcoin. | THE TUTORIAL |
| 3 | Prove a domain event and drive real application logic from it. | GOOD |
| 4 | The proven event carries policy, enforced against a second independent proof system. | PROOFSETTLE |

The plain version: everyone else uses the oracle to **look something up**. This uses it to **carry an instruction**, and the instruction is the buyer's own.

Levels 1 to 3 all treat the oracle as a source of facts. At level 4 the contract holds no opinion of its own: it reads `requiredMeasurement` out of the proven Sepolia event and enforces exactly that. **Change what the buyer asked for and the settlement rule changes with it, with no redeploy and no admin key.**

The data never touches either chain

IN AMARA'S BROWSER

The record is sealed to a key that exists only inside the enclave

X25519, HKDF, AES-GCM in thirty lines of WebCrypto. Only keccak256 of the record goes on chain.

INSIDE THE PAYMENT

The sealed bytes ride behind the calldata arguments

The ABI decoder ignores what follows the arguments. Two forge tests pin that against the compiled bytecode. No contract changed.

INSIDE THE ENCLAVE

Open, check the commitment, run the model the buyer named

The model is in the image; its hash is in the attestation's identity. Any other model, or a record that does not match, is refused and signed as Rejected.

INSIDE THE SETTLEMENT

The answer rides back, sealed to Amara's one-time key

She opens it and checks it hashes to the resultHash the contract recorded. The worker that carried it cannot read it.

The keys are part of what Google signs

The enclave asks the Confidential Space launcher for its attestation token with its own two public keys as nonces. So the token does not only say which image ran. It says **these keys were generated inside it**, and the page checks for that.

Zero runtime dependencies inside the boundary

Keccak, secp256k1, the envelope and the model are a few hundred lines of plain Node, checked against ethers outside the enclave. A supply chain is a poor thing to put inside a trust boundary.

LIVE TODAY

A real job, paid on Sepolia, settled on Creditcoin

Payment	0xcc8283fc167cf6ad4ce66b19b20e84a3640cb10c191cbdb58b461f6c978ac1ce Sepolia, block 11,673,891, carrying the applicant's record sealed to the enclave
Settlement	0xef8191d1e33a8963e6c3d9c80d661b2159b10d0d28f08c93cf4628d51d166eb2 Creditcoin CC3, 486,430 gas, 0.000243 CTC, carrying the answer sealed to the buyer
Settlement contract	0x010C1F801d5FAE37FD43C9B21025fE579dCC1F06
Enclave registry	0xAc014b7Df7f9b2dA343d895F22a392c655338bFb
Escrow, Sepolia	0xaF89A479E20890fDfFa4DeaDd3b5A2f7957b1E40
Signed by	An attested Confidential Space enclave on AMD SEV, secure boot on, debug disabled since boot. Its measurement is the container image digest, 0xea75311d...91b741, and the Google-signed token that says so, with both enclave keys in its nonces, is committed in the repository.

Four lines, in this order, and together they are the argument

1. TransactionVerified from the Attestcoin verifier precompile itself. The protocol, not this project, attesting the payment is real.
2. ProofConsumed, marking that query spent so it cannot settle twice.
3. JobSettled, carrying the enclave the buyer's own policy permitted.
4. Transfer, minting exactly what was locked and nothing more.

And here it is refusing

A job where the buyer demanded the **previous** enclave build, revoked that morning with the reason on chain. The live enclave answered anyway. The contract rejected it, and nothing was released.

Transaction	0xaab7768b...ccb87a Creditcoin CC3, block 5,462,957, status 0, 416,682 gas
Revert	EnclaveNotAccepted(0xc0a8e85a..., 0x986fb1b9...) The build the buyer named, and the one that actually signed. The error names both halves of the disagreement rather than failing silently.
Revocation	0xa1d4eb78...058c9 "superseded by proofsettle-enclave/1.2.0: model inside the image, sealed input and output, keys generated inside the enclave"

The build it refused was the newer, better one

The buyer asked for 1.1.0. The 1.2.0 enclave, attested, nonce-bound, carrying the model, did the work and signed, and was refused for not being what was asked for.

The contract has no opinion about which build is better. It enforces the one the buyer named. That is what makes the policy theirs rather than mine. An earlier refusal, on 24 August, went the other way: an attested build refused because the buyer had named an unattested one.

A rail that only ever pays is not a rail. The refusal is the load-bearing half, so it is mined, public, and reproducible from the repository in one command.

Buy one decision, and watch both proofs land

`proofsettle.pages.dev/desk.html` is the buyer's side. Pick an applicant, name the build you will accept, pay once on Sepolia. The page then reads every step off the two chains and opens the answer with a key only it holds.

✓ Provider	Registry binding read live; Google's signature on the attestation checked in the browser; the enclave's keys found in the token's nonces
✓ Paid	Mined on Sepolia, with the sealed record inside the same transaction
✓ Attested	Height read from the ChainInfo precompile at 0x...0fd3
✓ Settled	Both proofs in one transaction, the answer opened and its hash matched against the chain
✗ Or refused	Demand the revoked build and the page finds the reverted settlement and decodes <code>EnclaveNotAccepted</code> by name

approve

Score **75.51%**, from `proofsettle-enclave/1.2.0`. Only this browser can read this.

supplier on time		+0.30
no defaults		+0.27
inflow regularity		+0.20
balance volatility		+0.16
months of history		-0.09
no loans yet		-0.17

The thin-file applicant Amara exists for. Six months of records, no loans, suppliers paid on time. Weighted towards behaviour, not length of history.

MEASURED, NOT ASSERTED

Measuring the ground before building on it

284

contracts called the verifier in 17.4 days, to 10 September

92.2%

of that traffic came from just two of them

260/284

have exactly one sending address

15

senders on the most-used contract, which is the tutorial

Before building on a protocol it is worth knowing how it is used, so I measured it rather than guessing, twice: every event the verifier precompile emitted on CC3 testnet across 100,000 blocks, complete coverage, no sampling, reproducible by anyone with `script/measure.ts`.

The final fortnight doubled the contracts and quadrupled the traffic. The pattern is what a testnet in its testing phase looks like, and it shaped this design: **build proofs locally, with the hosted service as the fallback**, and read the chain rather than assume it.



A green test suite is not evidence

51

contract tests, every guarantee paired with a negative case, plus enclave, worker and three browser suites

11/11

mutations killed. Each removes one guarantee and proves the test defending it goes red

10

stages in `npm run judge:verify`, ending with live readbacks from Creditcoin. No key, no account

The test that matters most

Anyone can deploy their own escrow, emit a perfectly well formed payment event naming themselves as provider for any amount, and obtain a completely genuine Attestcoin proof of it. **The proof is real. The settlement would be theft.** Pinning the emitting contract is what makes a true proof also a relevant one.

Checked against the live network, not the docs

- ✓ A real proof verifies on chain: the precompile returns true
- ✓ A corrupted proof reverts with "Merkle proof validation failed"
- ✓ Sepolia is chain key 1, read from the ChainInfo precompile
- ✓ Attestation runs 7 to 9 minutes behind, measured, not quoted

What this does not do

No hardware attestation on chain

Checking the hardware's own certificate the way a browser checks a website's would cost far more gas than any settlement is worth. No smart contract does this today, and one that claims to is worth reading closely.

The registry records the binding, a hash of the attestation document and where to fetch it. The contract enforces the binding.

Anyone can recheck the evidence and disagree in public.

The rail is one-directional today

Attestcoin carries attested data into Creditcoin. Writability, which would carry a settlement receipt back out, is documented as still under third-party testing and is not on testnet, as the protocol team also said at the kickoff AMA.

So the return leg is a plainly labelled conventional relayer, and the buyer's refund is gated on elapsed time rather than on a receipt.

Writability is the roadmap step that removes the last trusted component.

Escrow against verified delivery is a credit instrument

Creditcoin exists because credit cannot flow to people whose creditworthiness cannot be verified. ProofSettle is the same shape applied to work instead of to borrowers. **Amara is not an incidental example: the reason her borrowers cannot be scored is a compute problem, not a credit one.**

Lenders scoring thin files

Exactly Creditcoin's existing constituency, blocked on a step that happens off chain and cannot be checked.

Compute providers

Independent GPU and TEE operators who cannot prove they ran what they claim, and are paid on trust or not at all.

Agent-to-agent payments

An autonomous buyer cannot read a reputation score. It can read a signature from a named enclave build.

Where this goes after the hackathon

Shipped

Two-proof settlement live on CC3 testnet, with a public page anyone can verify against
Deep Attestcoin use that is not the tutorial

Next

Attestcoin writability closes the loop, replacing the conventional return leg
Removes the last trusted component

Next

Settle in an asset that already has value, through Creditcoin's own DEX
Real money in the existing economy, not a token this project invented

Then

More source chains as Attestcoin adds them, same contract
Creditcoin becomes the settlement layer for multi-chain compute

Then

A provider marketplace: enclave builds compete on measurements
A reason for compute operators to hold and spend CTC

The test that counts

One real lender, one real scoring model, mainnet
Everything above is scaffolding for this

Nothing here asks you to take my word for it

Every claim in this deck has something you can click or run. That is the point of building it this way, and it is the only reason any of it should be believed.

Click

The settlement, on Creditcoin's block explorer, with its four logs.

The payment, on Etherscan.

The refusal, a public transaction that failed for the right reason.

The page at proofsettle.pages.dev, which re-runs every check in your browser and verifies Google's attestation signature while it is there.

The repository, including the commit history that shows this happening.

Or run

```
forge test
```

```
./script/mutation-check.sh
```

```
node enclave/verify-token.mjs enclave/attestation.jwt
```

```
node site/test-attestation.mjs
```

```
npx tsx script/measure.ts
```

51 tests. 11 mutations, each removing one guarantee and proving the test defending it goes red. The attestation checked against Google's live keys. The protocol measurement reproduced from scratch.

None of this needs me present, a demo environment, or a video. It is on two public chains and a public page, and it will still be there after the judging.

IN ONE LINE

Most Attestcoin integrations ask the oracle for a fact.
This one asks it for a policy, and enforces that policy against a second,
unrelated cryptographic system.

So that Amara can buy an inference from someone she has no reason to trust, and pay only
when both proofs agree she got what she paid for.

Deployed on CC3 testnet

Sealed jobs settled end to end

51 tests, 11 mutations killed

Verifiable by anyone, in a browser